

11 *Use of the Data Booklet is relevant to this question.*

A student mixes 20.0 cm^3 of 5.00 mol dm^{-3} sulfuric acid with an equal volume of 5.00 mol dm^{-3} sodium hydroxide. The initial temperature of both solutions is 25.0°C .

The maximum temperature reached after the reaction is 50.0°C . Assume the density of both solutions is 1 g cm^{-3} .

What is the value for the enthalpy change of neutralisation calculated using these values?

- A** -4.18 kJ mol^{-1}
- B** -20.9 kJ mol^{-1}
- C** -41.8 kJ mol^{-1}
- D** -83.6 kJ mol^{-1}